1. Create a new file named lab5.txt using the vi/vim editor and enter at least five lines of text.

```
akansha@localhost:~$ vi lab5.txt

# Inside vi/vim editor:
# Press 'i' to enter Insert mode, then type 5 lines:
Welcome to Linux Administration Lab 5
This sheet covers vi/vim editor commands
Learning modes: Normal, Insert, and Command-Line mode
Practicing line editing, search, and substitution
Red Hat Enterprise Linux laboratory assignment

# Save and exit:
Press 'Esc', type ':wq', and press Enter.
```

2. Open an existing file named lab5.txt using vi/vim and display its contents.

```
akansha@localhost:~$ vi lab5.txt
akansha@localhost:~$ cat lab5.txt
Welcome to Linux Administration Lab 5
This sheet covers vi/vim editor commands
Learning modes: Normal, Insert, and Command-Line mode
Practicing line editing, search, and substitution
Red Hat Enterprise Linux laboratory assignment
```

3. Demonstrate switching between Normal mode, Insert mode and Command-Line mode in vi/vim.

```
akansha@localhost:~$ vi lab5.txt

1. Normal mode to Insert mode        : Press 'i' (insert before cursor) or
'a' (append after cursor)
2. Insert mode back to Normal mode   : Press 'Esc' key
3. Normal mode to Command-Line mode : Press ':' (colon appears at the bottom
left)
4. Exit Command-Line mode to Normal  : Press 'Esc' key twice
```

4. Insert new text before and after the current cursor position and verify the changes.

```
# Position cursor on a word in Normal mode:
- Press 'i' -> type 'START: ' -> press 'Esc' (Inserts text BEFORE cursor)
- Press 'a' -> type ' :END'   -> press 'Esc' (Inserts text AFTER cursor)

akansha@localhost:~$ cat lab5.txt
START: Welcome to Linux Administration Lab 5 :END
This sheet covers vi/vim editor commands
Learning modes: Normal, Insert, and Command-Line mode
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Red Hat Enterprise Linux laboratory assignment
```

5. Add a new line below the current line and another new line above the current line using vi/vim commands.

```
# In Normal mode:
- Press 'o' (small letter o) : Creates a new empty line BELOW the current line
and enters Insert mode.
- Press 'O' (capital letter O): Creates a new empty line ABOVE the current line
and enters Insert mode.

akansha@localhost:~$ cat lab5.txt
[Added line above using 'O']
START: Welcome to Linux Administration Lab 5 :END
[Added line below using 'o']
This sheet covers vi/vim editor commands
```

6. Save the current file without exiting vi/vim.

```
# In Normal mode:
Type ':w' and press Enter.

Bottom status bar output:
"lab5.txt" 7L, 245B written
```

7. Save the current file and exit vi/vim using the appropriate command.

```
# In Normal mode:
Type ':wq' or 'ZZ' and press Enter.

akansha@localhost:~$ ls -l lab5.txt
-rw-r--r--. 1 akansha akansha 245 Oct 31 11:20 lab5.txt
```

8. Exit vi/vim without saving the changes made to a file.

```
# In Normal mode:
Type ':q!' and press Enter (Forces exit, discarding all unsaved edits).
```

9. Move the cursor to the beginning and end of a line using vi/vim commands.

```
# In Normal mode:
- Press '0' (zero) or '^' : Moves cursor to the BEGINNING of the current line.
- Press '$' (dollar sign) : Moves cursor to the END of the current line.
```

10. Move the cursor to the beginning and end of the file using vi/vim commands.

```
# In Normal mode:
- Press 'gg' or ':1'      : Moves cursor to the BEGINNING of the file (Line 1).
- Press 'G' (Capital G)   : Moves cursor to the END of the file (Last Line).
```

11. Delete a character, a word and a complete line using appropriate vi/vim commands.

```
# In Normal mode:
- Delete a single character : Press 'x'
- Delete a word             : Press 'dw'
- Delete a complete line    : Press 'dd'

akansha@localhost:~$ cat lab5.txt
[Line deleted using 'dd']
```

12. Copy the current line and paste it below the current line.

```
# In Normal mode:
- Press 'yy' (yank/copy current line)
- Press 'p'  (paste copied line BELOW current line)

akansha@localhost:~$ cat lab5.txt
This sheet covers vi/vim editor commands
This sheet covers vi/vim editor commands
```

13. Copy multiple lines and paste them at another location in the file.

```
# In Normal mode:
- Press '3yy' (copies 3 consecutive lines starting from cursor)
- Move cursor to target position
- Press 'P' (capital P pastes ABOVE cursor) or 'p' (pastes BELOW cursor)
```

14. Search forward for a specified word in the file and move between matching occurrences.

```
# In Normal mode:
- Type '/Linux' and press Enter (Searches FORWARD for 'Linux')
- Press 'n' : Moves to NEXT occurrence forward
- Press 'N' : Moves to PREVIOUS occurrence backward
```

15. Search backward for a specified word in the file and move between matching occurrences.

```
# In Normal mode:
- Type '?mode' and press Enter (Searches BACKWARD for 'mode')
- Press 'n' : Moves to NEXT occurrence backward
- Press 'N' : Moves to PREVIOUS occurrence forward
```

16. Replace the first occurrence of a specified word on the current line using vi/vim.

```
# In Command-Line mode (Press ':'):
Type ':s/vi/vim/' and press Enter.

Before: This sheet covers vi editor commands
After : This sheet covers vim editor commands
```

17. Replace all occurrences of a specified word throughout the file using a suitable substitution command.

```
# In Command-Line mode (Press ':'):
Type ':%s/Linux/RHEL/g' and press Enter.

Bottom status bar output:
5 substitutions on 4 lines

akansha@localhost:~$ cat lab5.txt
Welcome to RHEL Administration Lab 5
This sheet covers vim editor commands
Learning modes: Normal, Insert, and Command-Line mode
Red Hat Enterprise RHEL laboratory assignment
```

18. Undo the last change and then redo the change using vi/vim commands.

```
# In Normal mode:
- Press 'u'      : Undo the last change
- Press 'Ctrl+r' : Redo the change that was undone
```

19. Use line numbers to display the current line number and jump directly to a specified line.

```
# In Command-Line mode (Press ':'):
- Display line numbers         : Type ':set nu' and press Enter
- Hide line numbers            : Type ':set nonu' and press Enter
- Jump directly to line number : Type ':3' or '3G' in Normal mode (Jumps to Line
3)
```

20. Create, edit, search, replace, copy, paste, save and exit a text file using vi/vim as a complete workflow.

```
akansha@localhost:~$ vi workflow.txt
# Step 1: Press 'i' -> Type initial content -> Press 'Esc'
# Step 2: Press 'o' -> Add new line -> Press 'Esc'
# Step 3: Type '/Linux' to search -> Press 'n'
# Step 4: Type ':%s/Linux/RHEL/g' to replace all occurrences
# Step 5: Press 'yy' on line 1 -> Press 'p' to duplicate
# Step 6: Type ':wq' to save and exit

akansha@localhost:~$ cat workflow.txt
RHEL System Administration Lab
RHEL System Administration Lab
Complete vi/vim workflow executed successfully
```